Md Shaqib Hussain

AI/ML Engineer

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Professional Summary

AI/ML Engineer with a Post Graduate Diploma in Big Data Analytics (AI & ML) from C-DAC and a B.Tech in Computer Science Engineering, with hands-on experience in machine learning, deep learning, and Generative AI. Skilled in Python, SQL, Scikit-learn, TensorFlow, Keras, PySpark, LLMs, RAG, and LangChain, with experience developing classification, forecasting, and AI-driven solutions. Familiar with FastAPI, Streamlit, and AWS for deploying machine learning applications. GATE 2025 qualifier with strong analytical and problem-solving skills, seeking to build scalable AI/ML solutions for real-world applications.

Technical Skills

Statistical Analysis: Statistics, Probability, Hypothesis Testing, A/B Testing, Statistical Modeling, Time Series Analysis, Statsmodels

Programming & Data Analysis: Python, SQL, Pandas, NumPy, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA)

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Ensemble Learning, Random Forest, XGBoost, LightGBM, Gradient Boosting

Deep Learning: TensorFlow, Keras, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), LSTM, Transfer Learning, Transformers, Attention Mechanisms, Model Fine-Tuning

Generative AI & NLP: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, NLP, Text Classification, Embeddings, Vector Databases

Data Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, Microsoft Excel (Advanced)

Big Data: Apache Spark, PySpark, Hadoop

Databases: MySQL, SQL Joins, Aggregations, Window Functions, Data Warehousing

Deployment & Cloud: FastAPI, Streamlit, Docker, REST APIs, AWS EC2, AWS S3, Linux, Shell Scripting

Tools: Git, GitHub, VS Code, JupyterLab

Experience

Graduate Engineer Trainee – Data Analyst **Mar 2026 – Aug 2026**
SINCLUS Engineering & Consulting Pvt. Ltd. (BPCL Project) *Mumbai, India*

- Queried and consolidated operational data from multiple source systems using SQL, reducing manual reporting turnaround by an estimated 30% for the analytics team.
- Performed statistical analysis and exploratory data analysis (EDA) on production datasets to identify trends and anomalies, supporting weekly business review decisions.
- Built and maintained Power BI dashboards tracking 10+ key operational metrics, improving visibility for cross-functional stakeholders across 3 departments.
- Automated recurring data extraction and cleaning workflows in Python and Excel, cutting manual processing time and reducing reporting errors.
- Partnered with cross-functional teams to translate business requirements into analytical deliverables, presenting findings to senior stakeholders.

Projects

IntelliScan: AI-Assisted Cancer Screening System | *TensorFlow, Keras, CNN, Transfer Learning, Scikit-learn* **Feb 2026**

- Designed and trained a CNN-based deep learning model using TensorFlow and Keras for automated binary cancer detection and multi-class classification across 5+ cancer categories.
- Applied transfer learning (VGG-style architecture) and systematic hyperparameter tuning to improve model generalization on imbalanced classes, achieving 91% validation accuracy and 0.93 AUC-ROC.
- Implemented data augmentation, dropout regularization, and early stopping to prevent overfitting; evaluated outputs using precision, recall, F1-score, and confusion matrices – lifting classification accuracy by 18%.
- Leveraged Apache Spark (PySpark) for distributed preprocessing and feature extraction on 50,000+ images, enabling reproducible and scalable model training workflows.

Air Quality Forecasting System | *Python, Scikit-learn, LightGBM, XGBoost, Statsmodels, Streamlit, AWS EC2* **Jan 2026**

- Built an end-to-end ML pipeline for multi-hour AQI forecasting – covering data ingestion, exploratory data analysis (EDA), time-series feature engineering, model training, evaluation, and AWS deployment.
- Engineered lag variables, rolling statistics, and interaction features using Pandas and Statsmodels to capture complex temporal dependencies across 5+ air pollutant types.
- Trained and cross-validated LightGBM and XGBoost ensemble models across 10+ forecast horizons, achieving R^2 of 0.89 and RMSE of 12.4 – a 22% improvement over baseline.
- Deployed an interactive Streamlit application on AWS EC2 enabling real-time AQI forecasts from user-uploaded CSV data, validated by 20+ test users during evaluation.

Education

Post Graduate Diploma in Big Data Analytics (AI & ML) — **Percentage: 86.88%**
CDAC – Centre for Development of Advanced Computing, Pune — *Aug 2025 – Feb 2026*

Bachelor of Technology in Computer Science and Engineering — **CGPA: 7.54**
Heritage Institute of Technology, Kolkata — *Aug 2021 – Jun 2025*

Certifications & Academic Highlights

GATE 2025 – Computer Science and Engineering (Qualified) — **Mar 2025**

CDAC PG Diploma in Big Data Analytics (AI & ML) – 86.88% — **Feb 2026**

Extracurricular Activities

National Service Scheme (NSS) — **Aug 2021 – Jun 2025**
Coordinator – Heritage Institute of Technology, Kolkata

- Led campus and community awareness drives; coordinated educational support programs in local schools in Kolkata, benefiting 100+ students over 4 years.